Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

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Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and they struggle on knowledge-intensive tasks. In this work, we explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation.

1. Methods Overview

RAG models combine a pre-trained sequence-to-sequence model (parametric memory) with a dense vector index of Wikipedia (non-parametric memory) accessed via a neural retriever. We leverage Dense Passage Retrieval (DPR) to provide latent document retrieval, and BART as the generator.

2. RAG Architectures: RAG-Sequence vs. RAG-Token
We propose two formulations of RAG models that differ in how the retrieved documents are marginalized across generated output tokens:

RAG-Sequence Model: The RAG-Sequence model uses the same retrieved document to generate the complete sequence. Formally, it treats the retrieved document as a single latent variable that is marginalized to get the seq2seq probability p(y|x) via a top-K approximation:
*p_RAG-Sequence(y|x) = sum_{z in top-k(p(z|x))} p_eta(z|x) * prod_{i=1}^N p_theta(y_i | x, z, y_{1:i-1})*

RAG-Token Model: The RAG-Token model can draw from different latent documents for each target token. This allows the generator to synthesize content from multiple documents when generating an answer:
*p_RAG-Token(y|x) = prod_{i=1}^N sum_{z in top-k(p(z|x))} p_eta(z|x) * p_theta(y_i | x, z, y_{1:i-1})*

Non-Parametric Retriever (DPR): The retrieval component p_eta(z|x) is based on Dense Passage Retrieval. It uses a bi-encoder architecture: a query encoder and a document encoder, both based on BERT-base, trained on question-answer pairs.

3. Empirical Results & Groundedness
We evaluate RAG on open-domain question answering benchmarks including Natural Questions, TriviaQA, WebQuestions, and CuratedTREC, as well as knowledge-intensive generation tasks such as Jeopardy question generation and MS-MARCO.

Key Findings:
1. RAG models achieve state-of-the-art results on Open-Domain QA, outperforming parametric-only models like T5-11B while using significantly fewer parameters.
2. **Factual Grounding and Reduced Hallucination:** Because generation is conditioned on retrieved non-parametric text passages, RAG produces answers that are significantly more factual, specific, and verifiable compared to standard Seq2Seq models.
3. **Knowledge Updating:** Non-parametric memory can be updated simply by swapping or updating the document index without requiring expensive retraining of the generator network.